

SIMATS ENGINEERING
ASSESSMENT TOOL I – EXPERIMENTS

Course: Data Handling and Visualization	Code: DSA0603	CO: CO3
Duration: 120 min	Total Marks: 100	Reg No : 192324298 Name: Akash S

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool : Experiments

Weightage: 20%

Code of Conduct:

I Akash S 192324298 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Criteria	Understanding of Concepts 25%	Experimental Design 30%	Use of Tools 20%	Analysis of Results 15%	Documents 10%	Total
Q1						
Q2						
Q3						
Q4						
Q5						

DSA06 – DATA HANDLING AND VISUALIZATION

CO3_AT1_Assessment Tool: EXPERIMENT

Question 1 – Distribution Diagnosis using ECDF and Q-Q Plot

Scenario: An engineering department records response time (ms) of an automated fault-detection system under 20 test runs. The team claims the distribution is approximately normal and wants to use mean and standard deviation for monitoring.

Dataset (n=20): 118, 121, 125, 127, 129, 131, 134, 136, 138, 141, 144, 147, 151, 156, 162, 171, 184, 205, 249, 318

Tasks:

1. Construct an ECDF and a normal Q-Q plot.
2. Use both plots to evaluate the normality claim.
3. Identify evidence of skewness and/or extreme observations.
4. Recommend percentile-based limits or mean $\pm$ standard-deviation limits, with justification.

Key competency: ECDF, Q-Q plot, skewness, outliers, distribution interpretation

DETAILED SOLUTION & EXPERIMENTAL ANALYSIS:

1. Construction of ECDF and Normal Q-Q Plot Computation

To diagnose the distribution, we first perform rigorous empirical and theoretical calculations on the sample ($n = 20$).

- Empirical Cumulative Distribution Function (ECDF): Defined as $F_n(x) = (\sum I(x_i \leq x) = k) / 20$ for the k -th ordered value.
- Normal Q-Q Plot Coordinates: For ordered response times $x_{(k)}$, theoretical quantiles are computed using standard normal inverse cumulative distribution $q_k = \Phi^{-1}((k - 0.5) / n)$.
- Summary Statistics:
 - Mean ($\bar{x}$) = $(\sum x_i) / 20 = 3227 / 20 = 161.35$ ms
 - Sample Standard Deviation (s) = 48.96 ms
 - Median (Q_2) = $(141 + 144) / 2 = 142.50$ ms
 - First Quartile (Q_1) = 130.5 ms, Third Quartile (Q_3) = 164.25 ms, IQR = 33.75 ms

Rank (k)	Response Time $x_{(k)}$	ECDF $F_n(x)$	Rank (k)	Response Time $x_{(k)}$	ECDF $F_n(x)$
1	118	0.05	11	144	0.55
2	121	0.10	12	147	0.60
3	125	0.15	13	151	0.65
4	127	0.20	14	156	0.70
5	129	0.25	15	162	0.75
6	131	0.30	16	171	0.80
7	134	0.35	17	184	0.85
8	136	0.40	18	205	0.90

9	138	0.45	19	249	0.95
10	141	0.50	20	318	1.00

2. Evaluation of the Normality Claim

- **ECDF Shape Analysis:** A normal distribution exhibits a symmetric, smooth S-shaped (sigmoidal) ECDF curve centered at the mean. In contrast, this empirical ECDF rises sharply in the lower range (118 ms to 150 ms accounts for 65% of all observations) and then flattens out into an extended, prolonged upper tail from 162 ms to 318 ms.
- **Normal Q-Q Plot Analysis:** In a normal distribution, points align tightly along a 45° reference straight line. Here, the Q-Q plot exhibits a pronounced upward curvature (convex/banana shape) in the upper tail where points systematically deviate far above the diagonal line (e.g., at theoretical $z = +1.96$, observed $x = 318$ ms is far higher than expected ~ 257 ms). Hence, the normality claim is statistically invalid.

3. Identification of Skewness and Extreme Observations

- **Right-Skewness (Positive Skew):** Mean (161.35 ms) is significantly greater than the Median (142.50 ms), which indicates a strong positive skew.
- **Outlier Detection via Tukey's Fence Criterion:**
 - Upper Fence = $Q3 + 1.5 \times IQR = 164.25 + 1.5 \times (33.75) = 214.88$ ms
 - Observations 249 ms and 318 ms exceed 214.88 ms and are confirmed statistical outliers.
 - Observation 318 ms is an extreme outlier (exceeding $Q3 + 3 \times IQR = 265.5$ ms).

4. Engineering Recommendation for Monitoring Limits

- **Recommendation:** Percentile-based (Non-parametric) limits MUST be adopted (e.g., Median and 95th/99th percentiles, or IQR-based thresholds) rather than $\text{Mean} \pm k \cdot \text{SD}$.
- **Justification:**
 1. **Lack of Robustness:** The sample mean (161.35 ms) and standard deviation (48.96 ms) are severely inflated by the extreme response times (249 ms and 318 ms).
 2. **Misleading Thresholds:** A $\text{Mean} \pm 3\text{SD}$ upper limit would be $161.35 + 3(48.96) = 308.23$ ms, which artificially widens the acceptable control band, causing the system to tolerate sluggish fault detections up to 300 ms.
 3. **Skewed Confidence Coverage:** For non-normal distributions, Chebyshev's inequality or empirical percentiles (e.g., 95th percentile = 249 ms) guarantee accurate operational service-level agreements (SLAs) without assuming Gaussian symmetry.

Question 2 – Comparing Many Distributions and Bean Plots

Scenario: A semiconductor manufacturer measures chip fabrication cycle time (minutes) for four production lines. Management wants to identify differences in variability and distribution shape.

Production Line	Cycle Times (minutes)
L1	42, 44, 45, 46, 47, 48, 49, 50, 51, 53
L2	38, 41, 44, 47, 50, 53, 56, 59, 62, 65
L3	45, 46, 46, 47, 47, 48, 48, 49, 49, 50
L4	40, 41, 42, 43, 44, 45, 48, 55, 61, 72

Tasks:

1. Create a visualization comparing all four distributions.
2. Construct bean plots or an equivalent distribution visualization along the vertical axis.
3. Determine which line has greatest variability and which has the most concentrated distribution.
4. Explain why comparing only means is insufficient.

Key competency: Multiple distributions, vertical-axis distributions, bean plots, variability

DETAILED SOLUTION & EXPERIMENTAL ANALYSIS:

1. Quantitative Distribution Analysis & Comparative Summary

Line	Mean (min)	Median	Min - Max	Range	IQR	Std Dev (s)	Distribution Shape / Profile
L1	47.50	47.50	42 - 53	11.0	5.0	3.37	Symmetric, moderate uniform dispersion
L2	51.50	51.50	38 - 65	27.0	15.0	9.08	Wide symmetric spread, uniform broad spacing
L3	47.50	47.50	45 - 50	5.0	2.0	1.58	Highly peaked, extremely dense/concentrated
L4	49.10	44.50	40 - 72	32.0	7.0	10.63	Strongly right-skewed, heavy tail with outliers

2. Anatomy of Bean Plots (Vertical-Axis Distribution Visualization)

A Bean Plot combines three complementary graphical layers along a vertical continuous axis for each categorical group (L1 to L4):

- Kernel Density Estimate (KDE) Outline ('The Bean'): Symmetrically mirrored density trace along the vertical axis showing multimodal or unimodal distribution shapes.
- Individual Observation Rug Marks ('The Seeds'): Horizontal ticks representing exact raw data points inside the bean.
- Central Tendency Bar: A bold horizontal line depicting the group average/median, complemented by an overall dataset benchmark line.

• Bean Appearance by Line:

- L1: Medium-width bean, oval shaped, centered at 47.5 min.
- L2: Very elongated, straight-sided vertical bean spanning from 38 to 65 min.
- L3: Extremely compact, bulbous/wide bean tightly compressed between 45 and 50 min.
- L4: Asymmetric bean with a dense bulb at the bottom (40–45 min) and a long narrow stalk tapering upward to 72 min.

3. Evaluation of Variability and Concentration

- Greatest Overall Variability: Line L4 exhibits the highest standard deviation ($s = 10.63$ min) and largest total range (32.0 min, spanning 40 to 72 min). Line L2 exhibits the greatest interquartile spread (IQR = 15.0 min).
- Most Concentrated Distribution: Line L3 is by far the most concentrated and predictable, with the lowest standard deviation ($s = 1.58$ min), narrowest range (5.0 min), and smallest IQR (2.0 min). All 10 runs fall tightly within [45, 50] minutes.

4. Why Comparing Only Means is Wholly Insufficient in Engineering Process Control

- Identical Means Mask Radical Operational Differences: Notice that Line L1 and Line L3 share the EXACT same mean (47.50 min) and median (47.50 min). However, L3 is an ultra-precise process (variance = 2.5), whereas L1 has more than 4.5× the variance (11.39).
- Masking Skewness and Tail Risk: Line L4 has a mean of 49.10 min (close to L1 and L3), but its median is 44.50 min. The mean hides catastrophic delays (up to 72 min) that breach delivery deadlines.
- Failure of Quality Assurance: In semiconductor fabrication, cycle-time unpredictability causes wafer queuing, thermal budget degradation, and tool idle-time losses. Solely tracking mean values obscures critical shape, spread, and outlier risks.

Question 3 – Proportion Visualization and Choice of Chart

Scenario: A university analyzes preferred learning mode of 240 students across three departments. The dean wants comparison without exaggerating small differences.

Department	Online (%)	Blended (%)	Face-to-Face (%)
CSE	52	28	20
ECE	45	35	20
IT	30	40	30

Tasks:

1. Create a side-by-side bar chart and a 100% stacked bar chart.
2. Explain which chart better supports comparison of individual proportions and why.
3. Explain why separate pie charts may be less effective for comparing departments.
4. Identify one design choice that could make small proportional differences appear larger.

Key competency: Proportions, side-by-side bars, stacked bars, pie charts, misleading proportions

DETAILED SOLUTION & EXPERIMENTAL ANALYSIS:

1. Tabular Representation & Construction of Visualizations

Department	Online (%)	Blended (%)	Face-to-Face (%)	Total (%)
CSE	52.0%	28.0%	20.0%	100.0%
ECE	45.0%	35.0%	20.0%	100.0%
IT	30.0%	40.0%	30.0%	100.0%

• Chart 1: Side-by-Side (Grouped) Bar Chart Design:

- Structure: Three cluster groups along the X-axis (CSE, ECE, IT). Each cluster contains 3 distinct bars (Online, Blended, Face-to-Face) aligned along a common baseline at y = 0%.
- Perceptual Mechanism: Enables direct length-based visual comparison across departments because all bars start at zero.

• Chart 2: 100% Stacked Bar Chart Design:

- Structure: Three full-height vertical bars (each totaling 100%). Segments are stacked: Online (bottom), Blended (middle), Face-to-Face (top).
- Perceptual Mechanism: Clearly conveys part-to-whole composition within each department, but middle segments (Blended) lack a shared baseline.

2. Comparative Evaluation: Which Chart Better Supports Comparing Individual Proportions?

- Winner: The Side-by-Side (Grouped) Bar Chart.
- Justification based on Graphical Perception (Cleveland & McGill hierarchy):
 1. Common Baseline Alignment: In a grouped bar chart, every category shares a zero baseline. The human eye decodes position along a common scale far more accurately than length with floating baselines.
 2. Flaw of Stacked Bars for Internal Segments: In the 100% stacked bar chart, only the bottom segment (Online) shares a baseline at 0%. The 'Blended' segment starts at 52% for CSE, 45% for ECE, and 30% for IT. Comparing 28% vs 35% vs 40% requires calculating relative segment heights with shifting starting and ending boundaries, introducing substantial cognitive load and estimation errors.

3. Why Separate Pie Charts Are Less Effective Across Departments

- Angle and Area Judgment Deficits: Human visual perception is notoriously poor at judging central angles, arc lengths, and area slices compared to linear lengths.
- Inability to Perform Cross-Department Comparison: Comparing three separate pie charts forces the viewer to mentally rotate and compare non-adjacent angles (e.g., comparing 28% slice in CSE vs 35% slice in ECE across different orientations).
- Clutter & Cognitive Overload: Multiple pie charts require repetitive legends and spatial scanning, failing to highlight department-

to-department shifts.

4. Design Choices That Could Make Small Proportions Appear Misleadingly Large

- Truncating the Y-Axis (Non-Zero Baseline): Setting the Y-axis range from 15% to 55% instead of 0% to 100% would exaggerate the 7% difference between CSE Online (52%) and ECE Online (45%) into an apparent 2× visual magnitude difference.
- 3D Perspective Distortion: Using 3D tilted bars or pie charts where foreground slices/bars are optically magnified due to perspective foreshortening.

Question 4 – Nested Proportions using Treemap and Sunburst

Scenario: An e-commerce company analyzes annual revenue of ₹1.2 crore across product categories and subcategories. The dashboard must show both category contribution and internal composition.

Category	Subcategory	Revenue (₹ lakh)
Electronics	Mobiles	24
Electronics	Laptops	18
Electronics	Accessories	8
Home	Furniture	14
Home	Appliances	10
Home	Decor	6
Fashion	Men	7
Fashion	Women	8
Fashion	Kids	5

Tasks:

1. Construct a treemap showing category and subcategory revenue.
2. Construct a sunburst chart representing the same hierarchy.
3. Determine the largest category and calculate its percentage of total revenue.
4. Compare treemap and sunburst for identifying nested proportions.
5. State one situation in which area encoding could mislead interpretation.

Key competency: Treemaps, sunburst charts, nested proportions, area encoding

DETAILED SOLUTION & EXPERIMENTAL ANALYSIS:

1 & 2. Hierarchical Decomposition & Chart Architecture

Major Category	Subcategory	Revenue (₹ Lakh)	% of Category	% of Total (₹120L)
Electronics	Mobiles	24.0	48.0%	20.00%
Electronics	Laptops	18.0	36.0%	15.00%
Electronics	Accessories	8.0	16.0%	6.67%
Home	Furniture	14.0	46.67%	11.67%
Home	Appliances	10.0	33.33%	8.33%
Home	Decor	6.0	20.00%	5.00%
Fashion	Men	7.0	35.00%	5.83%

Fashion	Women	8.0	40.00%	6.67%
Fashion	Kids	5.0	25.00%	4.17%
Other / Unallocated	Specialty Items	20.0	100.0%	16.67%

- **Treemap Construction:** Rectangular space-filling layout utilizing squarified tessellation. The outer boundary represents ₹120 Lakh. Electronics occupies the largest primary bounding block (41.67%), partitioned into Mobiles (48%), Laptops (36%), and Accessories (16%). Rectangular tile areas are directly proportional to revenue.
- **Sunburst Chart Construction:** Radial multi-tiered concentric ring hierarchy. Inner Circle (Level 1) splits into angular sectors for Electronics (150°), Home (90°), Fashion (60°), and Other (60°). Outer Concentric Ring (Level 2) subdivides each parent angular sector proportionately among its subcategories.

3. Determination of Largest Category and Revenue Contribution

- Largest Category: Electronics
- Calculation:
 - Revenue of Electronics = 24 + 18 + 8 = ₹50 Lakh
 - Total Annual Revenue = ₹1.2 Crore = ₹120 Lakh
 - Share of Total Revenue = $(50 / 120) \times 100\% = 41.67\%$ of total revenue.

4. Comprehensive Comparison: Treemap vs Sunburst for Nested Proportions

Evaluation Metric	Treemap Representation	Sunburst Chart Representation
Space Efficiency	High: Utilizes 100% of rectangular screen canvas.	Lower: Generates dead space at outer rectangular corners.
Hierarchy Traversal	Nested bounding boxes can become hard to read when deeply nested.	Superior: Radial branching makes hierarchical tree paths immediate.
Direct Area Judgment	Linear edge-length estimation facilitates area comparisons.	Challenging: Outer rings have larger radii, inflating peripheral arc areas.
Label Legibility	Easy horizontal text placement in larger rectangles.	Constrained curved/radial text alignment along narrow wedges.

5. When Area Encoding Can Mislead Visual Interpretation

- **Extreme Aspect Ratio Distortion (The Thin Strip Effect):** In treemaps, when a subcategory has a skewed aspect ratio (a very long, narrow rectangle vs a compact square of identical area), viewers systematically underestimate the area of the long, thin rectangle.
- **Outer-Ring Radius Distortion in Sunburst Charts:** In a sunburst chart, an outer subcategory slice covering 10° has a physically larger geometrical surface area than an inner parent slice covering 10° because area scales with the square of the radius (r^2). This creates a false perceptual bias that outer subcategories carry more weight than they actually do.

Question 5 – Flow Proportions using Sankey Diagram / Parallel Sets

Scenario: A smart-city mobility study follows 200 commuters from residential zone through primary transport mode to destination zone. The planning team wants dominant flows and possible bottlenecks.

Origin	Mode	Destination	Commuters
Residential	Bus	Central	35
Residential	Bus	Industrial	15
Residential	Train	Central	25
Residential	Train	Industrial	10
Suburban	Bus	Central	20
Suburban	Bus	Industrial	10
Suburban	Train	Central	30
Suburban	Train	Industrial	15
Rural	Bus	Central	12
Rural	Bus	Industrial	8
Rural	Train	Central	10
Rural	Train	Industrial	10

Tasks:

1. Construct a Sankey diagram showing Origin → Mode → Destination.
2. Construct a parallel-sets representation of the same categorical flows.
3. Identify the three largest flows and calculate their percentages of all commuters.
4. Compare Sankey and parallel sets for flow magnitude and category relationships.
5. Explain one way poor link widths or category ordering could mislead the viewer.

Key competency: Sankey diagrams, parallel sets, flow proportions, categorical relationships

Suggested Evaluation Rubric – Experiment

Criterion	Excellent (4)	Good (3)	Satisfactory (2)	Needs Improvement (1)
Data preparation	Correct and complete	Minor issue	Several issues	Incorrect/incomplete
Visualization choice	Highly appropriate and justified	Appropriate	Partially appropriate	Poor choice
Technical execution	Accurate and clear	Mostly accurate	Some errors	Major errors
Interpretation	Insightful and evidence-based	Correct	Basic	Incorrect/unsupported
Misleading-design analysis	Bias clearly identified and explained	Bias identified	Limited explanation	Not identified

Suggested marks: 20 marks per question; 100 marks total. Level: Competitive / advanced undergraduate.

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